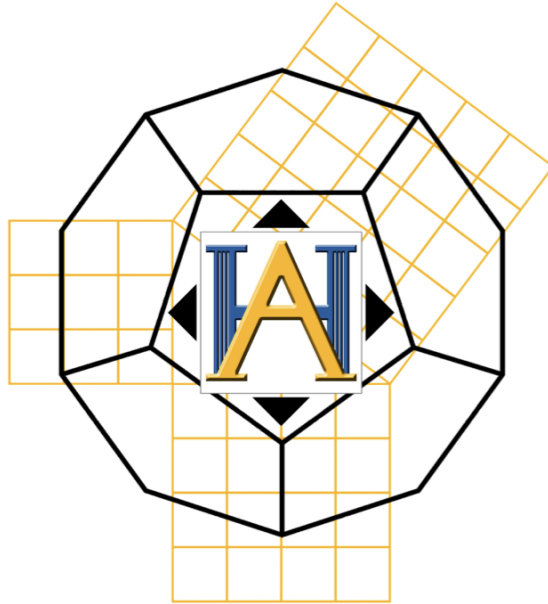




**AMERICAN HERITAGE SCHOOLS
COMPETITIVE MATHEMATICS**

2025 Matrices and Vectors Open Test

By Guilherme Salles

Rules

You will have 20 minutes to complete up to a maximum of 8 questions. You will get 1 point for every question right and 0 points for every question that is either wrong or blank. To answer a question, write and circle your answer next to that question's number. All answers must be simplified and rationalized.

Name: _____

1. Find the determinant of

$$\begin{bmatrix} -2 & 2 \\ 1 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & -10 & 1337 \\ 0 & -5 & 7 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -42 & 1 & 5 \\ -100 & 1 & 9 \end{bmatrix}$$

2. The eigenvalues of $\begin{bmatrix} 54 & 81 \\ 162 & 135 \end{bmatrix}$ are -27 and 216 . Find $\sqrt[3]{\begin{bmatrix} 54 & 81 \\ 162 & 135 \end{bmatrix}}$

3. Find the sum of the numbers next to the true statements. Assume that M is an arbitrary square matrix.

(2) For a nilpotent $n \times n$ matrix M , $M^k = 0$ for positive integer $k < n$

(3) For an idempotent matrix M , $I - M$ is also idempotent

(5) For a skew-symmetric matrix, M , of even order, $\det M = 0$

(7) $\sum_{k=1}^{\infty} M^k$ is defined if and only if the absolute value of the eigenvalues of M are less than 1

4. Let M be a 3×3 lower triangular matrix. Given that

$$\text{Tr} \left(M \begin{bmatrix} -3 & 0 & 0 \\ 0 & -12 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right) = 78$$

Find the minimum value of $\text{Tr}(M^2)$.

5. Define point $P = (7, 1, 6)$ and vector $\hat{v} = \langle 3, 5, 4 \rangle$ and P' as the reflection of P over $\hat{v}$. Find P' .

6. Find the reflection of the point $(1, 3)$ over $2x - 3y = 6$.

7. Suppose $M = \begin{bmatrix} x & -1 & 4 \\ 0 & y & -1 \\ 3 & 2 & z \end{bmatrix}$. If $x^2 + y^2 + z^2 = 1$, find the maximum value of the sum of the elements of M^2 .

8. Define $\text{adj}M$ to be the adjoint matrix of M . Let

$$M = \begin{bmatrix} 4 & 1 & -3 \\ 0 & 3 & 10 \\ 0 & 2 & 8 \end{bmatrix}$$

Find $\det(\text{adj}M)$.